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detecting and analyzing facial and eye features of drugs abuser using AI

Graduation Project

by

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Abstract

Drug addiction is a pervasive and deeply concerning issue, affecting millions of individuals worldwide. To detect whether a person is addicted, a blood test needs to be conducted. However, detecting drug addiction through a blood test is costly, takes time and resources and a long journey for each person to get the result. The journey starts with collecting a blood sample, typically through a needle inserted into a vein. The sample is then transported to a laboratory for analysis, where specialized techniques are employed. This duration can range from a few days to a week. To solve this issue, we will build a model called Drug Detection (DD) that detects drug users with minimal cost, effort, and time, only through an image of the face or eyes. The model analyses the size of the pupil of the eye or the difference between the two sides of the face to detect drug users. The DD system will detect drug addiction using deep learning, specifically by using different classifier approaches such as Convolutional Neural Network (CNN), Support vector machine (SVM), and structural similarity index measure (SSIM). To train this model, we collect data sets containing eye and face images of normal people and people who use stimulants. We aim to provide a high accuracy that achieves more than 90%. Lastly, the expected result of our project is to help police stations detect drug addiction with minimal time, effort, and cost.

Keywords: Drug Addiction – Deep Learning – Image Processing - Drug Detection.

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List of Abbreviations

DD	**Drugs detection**
CNS	**Control Nerves System**
AI	**Artificial Intelligence**
ML	**Machine Learning.**
SL	**supervised learning**
CNN	**Convolutional Neural Network.**
SVM	**support vector machine**
SSIM	**structural similarity index measure**
GAN	**generative adversarial network**
HOG	**Histogram of Oriented Gradients**
GIST	**Gastrointestinal stromal tumors**
LBP	**local Binary Pattern**
BISF	**Binarized Statistical Image Feature**
BOWs	**Bag of Word**
ROIs	**Region Of interest**
PUDUD	**people drugs use disordes**
SGD	**Stochastic gradient descent**
VGG	**Visual Geometry Group**
BN	**Bayesian Networ**

Chapter 1: Introduction

1.1 Introduction

Drug addiction is a significant issue affecting millions of individuals worldwide, and many governments fight the use of drugs. Illicit drugs are any plant or constructed substance that contains hypnotic or sedative elements, which, if used for purposes other than the medical purposes for which they are intended, infect the body with listlessness and lethargy and paralyze its activity. It also infects the central nervous, respiratory, and circulatory systems with chronic diseases and leads to a state of habituation. Alternatively, what is called "addiction" causes severe harm to mental, physical, and social health [1].

There is a physical change that appears on the face and eyes, the noticeable difference in the face is unbalancing the lips, and mandibular asymmetry [2], the meth effect on the nose is nose irritations and swells .and for cocaine, usually administered intranasally by 'snorting' [3]. Therefore, the use of stimulant substances can cause visible facial deformities and accelerate the biological ageing process, increasing the rate of facial changes by up to three to five times more than in ordinary people [2]. Furthermore, there is a pupil change. Regular pupil sizes are average in healthy adults, reacting in light and darkness [4]. However, taking illicit stimulant drugs can lead to changes in pupil size. Which is dilated, and there are no direct responses such as light and darkness responses [5]. This effect can last several hours while the drug affects the Central Nervous System (CNS) but goes away when the drug is removed from the body [6].

Traditional ways to detect drug abusers through methods such as urine drug testing and blood testing, the time it takes to detect drug use through traditional methods can vary; for example, urine drug testing typically provides results within a few days, while hair follicle testing can detect drug use over a longer time frame, up to several months. Therefore, the challenge is to detect the drug abuser in a short time [7] . Therefore, the challenge is to detect the drug abuser in a short time [7].Thus, our system provides a first and timely diagnosis of the problem of drug addiction. The development of a Drug Detection (DD) system that accurately and efficiently diagnoses drug users is crucial. This system uses deep learning techniques, specifically by using different classifier approaches such as Convolutional Neural

Network (CNN), Support vector machine (SVM), and structural similarity index measure (SSIM). By taking images of a person's eyes or face, the system can recognize and analyze these images and detect if the person in this image is a drug abuser or not through the size of the pupil of the eye or the difference between the two sides of the face. This technique helps prevent individuals who cannot control their awareness from engaging in harmful behaviors and allows for early intervention and support.

1.2 Motivation

Significant efforts and much money have been spent in Saudi Arabia to detect drug addiction in response to the control of drug addiction. The government has adopted thorough programs to detect and support those battling drug addiction. Therefore, it built specialized centers to detect early indicators of drug usage using various procedures such as blood tests. As previously said, these tests require effort and time to produce results. This motivated us to develop Drugs Detection (DD), a technology that accelerates and improves the existing method of detecting drug addiction. Drug detection (DD) can detect drug addiction by analyzing photos of eyes or faces and displaying the results in real-time with no effort.

1.3 Project Objectives

The objective of this project is to create a system that can identify drug users based on facial and eye analysis by using Deep learning techniques. The specific goals of the project include:

- To create a dataset.
- To build a model that is able to detect drug addiction.
- To design the system to process images of eye and face in real-time, allowing for quick identification of potential drug users.
- To determine the type of drugs.

1.4 Project Scope

The product of this project is an app that detects potential drug users Through the face and eyes. The site includes data collection, pre-processing, feature extraction, development of deep learning models, eye recognition algorithms, facial analysis algorithms, Real-time treatment, testing and evaluation to determine whether he is a drug user. Privacy will be processed, and reports will be submitted. The Application does not include Substance abuse intervention or treatment.

1.5 Project Timeline

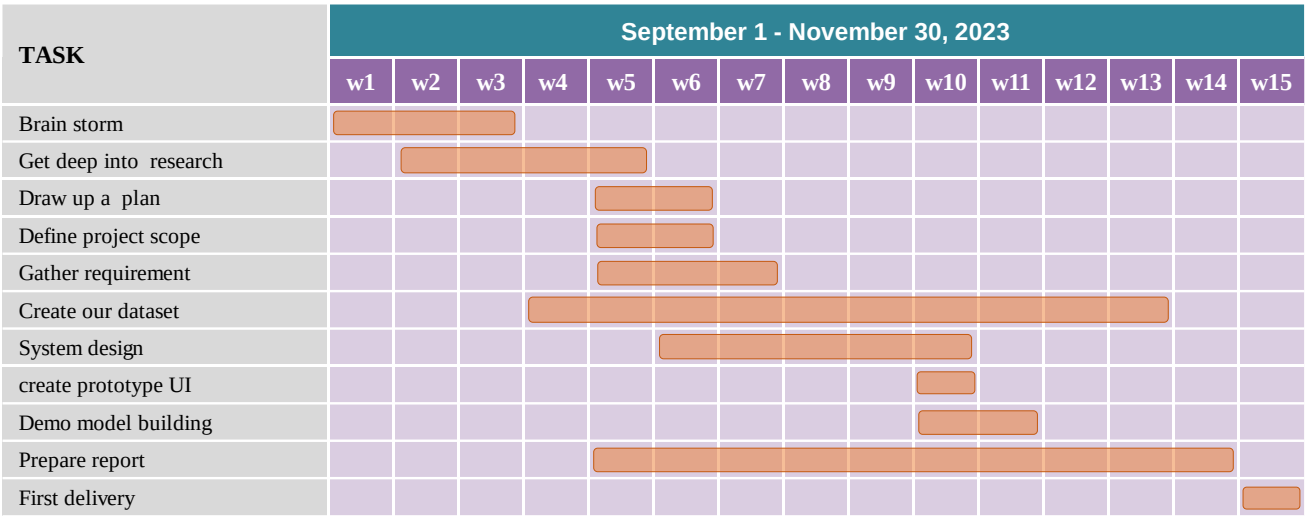


Figure 1-1 Gantt Chart

1.6 Document Organization

This project consists of:

- Chapter 1: introduction of the project, Problem Definition or motivation, Objectives, scope, and timeline.
- Chapter 2: Provides the reader with a summary of the literature review related to the application. and to understand the illicit drugs and their effect, we describe and summarize recent studies.
- Chapter 3: we introduce the system's requirements and describe each functional and non-functional requirement. The requirements specification is then described using a use case diagram. The last part of the chapter describes the development methodology and research methodology that we followed.
- Chapter 4: presents architectural design, object-oriented design, data modelling and user interface design.
- Chapter 5: implementation {2nd semester}
- Chapter 6: presents the conclusion, future work and the main goals that have been achieved.

Chapter 2: Literature Review

2.1 Introduction

Our project discusses how to detect drug addiction with minimal time, effort, and money by analyzing an image of a face or eye using deep learning. Therefore, in this chapter, we focus on the concept of understanding the drug effects on the face and pupil using deep learning, the focus is on the background of this field, the definitions, and the steps needed to achieve the goal of detecting the abuser drugs. We present the related work and existing similar systems and the effect of illicit drugs. The end summarizes the main points discussed in this chapter.

2.2 Background

Traditional methods of detecting drug abuse often rely on blood tests or subjective assessments [7]. Which may take a long time to intervene. Therefore, the project aims to take the benefit of advances in deep learning to develop an automated system that can analyze facial and eye features to identify potential drug users And the stimulant medications they use in real time. The project also holds of account privacy to ensure this Responsible and safe implementation. Ultimately, the project's goal is to contribute to Develop and promote effective tools and strategies in combating drug abuse Healthier and safer communities. In this section, we present the basic concepts that make up the basis of our work and what we will use in our system:

2.2.1 Type of Drugs

Drugs may be classified into five categories: depressants (e.g., alcohol), hallucinogens (e.g., cannabis), narcotics (e.g., heroin), stimulants (e.g., cocaine, methamphetamines), and other compounds (e.g., anabolic). In our project, we focus on stimulant drugs [8] [6].

2.2.1.1 Stimulants

Stimulants affect the central and peripheral nervous systems (CNS), resulting in different psychostimulant actions such as increased alertness, overbearing activity, and altered mood. Additionally, they can impact the cardiovascular and ocular systems, as well as disrupt temperature regulation. Higher doses and persistent use of these substances can cause acute psychosis [9].

There are different routes of administration, but the faster onset is after the drug is smoked or injected to be delivered to the brain quickly m, causing a euphoric rush. Because the effects are transient, repeated doses are often used to sustain the effect, leading to the binge-and-crash pattern common with stimulant misuse [10]. Because of the effect on CNS, there are outstanding changes such as dilated pupil and the effects that appear on the face in the long term, and we will harness machine learning to identify these changes.

2.2.2 changes with drugs

The most noticeable change in the face is the most prominent side effect of almost all, which makes it possible to detect illicit drug abusers only by observing the facial changes. The physiological side effect of meth and cocaine is dry mouth, which may lead to dental caries, and activation of mandibular muscles, which may lead to bruxism and, in some cases, tooth fracture. Because of the corrosive nature of the drug and broken teeth, which lead to unbalancing the lips and mandibular symmetry while increasing the chances for infections and face swells, in meth the effect on the nose is nose irritations and swells .and for cocaine, Usually administered intranasally by 'snorting,' pure cocaine is usually diluted by drug dealers with other substances the complications of nasal deformity have an incidence of more than 50% in users who snort cocaine [11] [3] [2]. Therefore, the use of stimulant substances can cause visible facial deformities and accelerate the biological ageing process, increasing the rate of changes by up to three to five times more than in ordinary people [2].

The eye and face are a rich source of biological and physiological information that can reveal the effects of drugs on the body [5], the anatomy of the eyeball contains three layers

(the fibrous layer, the vascular pigmented layer, and the nervous layer.), but we interact with the layer containing the (iris, pupil) which is a vascular pigmented layer [12] Figure 2-1.

The iris is a thin, contractile, pigmented diaphragm with a central pupil aperture. The iris's circular and radial muscles control the pupil size [13].

Regular pupil sizes are average in healthy adults, reacting to light and darkness. As for their reactions under bright lighting, their diameter is usually between 2 to 4 mm and may increase to 4 to 8 mm in the dark [4]. However, taking illicit stimulant drugs can lead to changes in pupil size. The dose must be at least 30mg to commence dilated pupil [14], leading to pupil dilation remaining relatively constant and not constricting in response to changes in lighting [5],such as methamphetamine and cocaine. This effect can last several hours while the drug affects the CNS Central Nervous System (CNS) [6] but goes away when the drug is removed from the body.

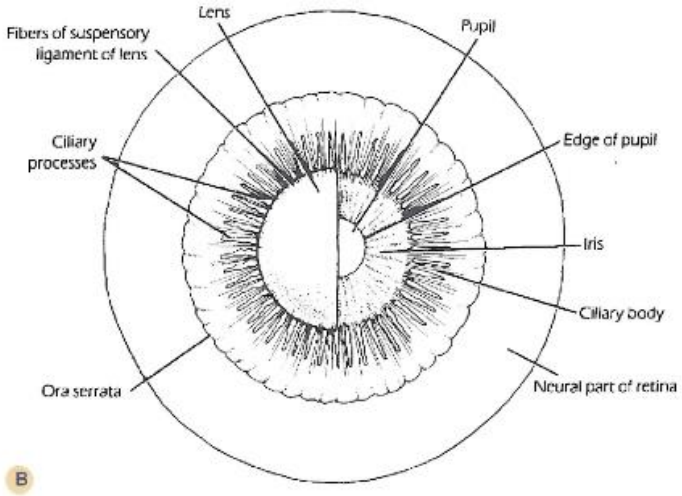


Figure 2-1 Anatomy of eye [12]

2.2.3 Artificial Intelligence

Artificial Intelligence (AI) is the part of computer science which is concerned with designing intelligent computer systems that exhibit the characteristics used to associate with intelligence in human behavior [15]. AI can be categorized into narrow AI, which is specialized in specific tasks, and general AI, which possesses human-like intelligence across various

domains. AI utilizes techniques like machine learning, deep learning, and natural language processing to process data, detect patterns, and improve performance [16].

2.2.3.1 Machine Learning

Machine learning [16] can be described as the process where a computer interacts autonomously with data, eliminating the need for intricate programming. This approach is fundamental in scenarios such as predicting outcomes using specific data and algorithms. It employs techniques like supervised and unsupervised learning. In supervised learning, the model is educated using labelled data sets, to classify or forecast future outcomes. On the other hand, unsupervised learning leverages unlabeled data to uncover concealed patterns within these data sets.

2.2.3.1.1 Supervised learning

Supervised learning (SL) [16] Figure 2-2, also known as supervised machine learning, is a subcategory of machine learning. Where the dataset consists of labelled data to train models to classify or predict output accurately. SL helps solve a variety real-world problem at scale, such as classifying spam in a separate folder from your inbox. SL uses a training set to train models to achieve the required output. The training dataset includes inputs and accurate outputs, which allow the model to learn over time. The algorithm measures its accuracy through the loss function, adjusting until the error is sufficiently reduced. Supervised learning can be classified into two types of problems:

- Problems: Classification uses an algorithm to accurately assign test data into specific categories. which means there are two classes such as Yes-No, True- false, etc. Some common classification algorithms are linear classifiers, and support vector machines (SVM).
- Regression is used to understand the relationship between dependent and independent variables. It is commonly used to make projections. Linear regression, logistical regression, and polynomial regression are popular regression algorithms.

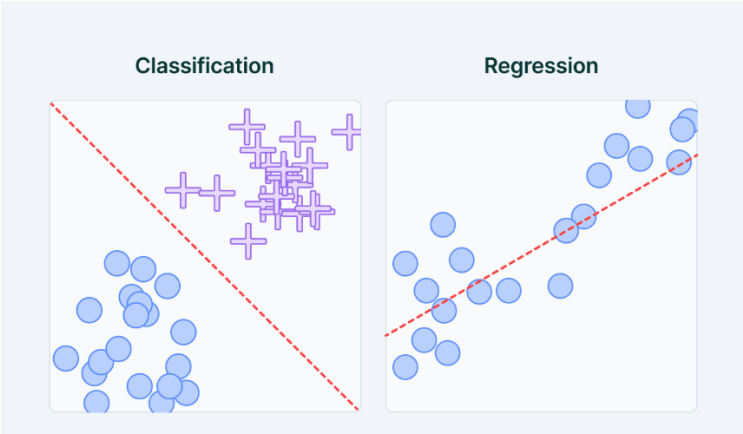


Figure 2-2 Supervised.

2.2.3.1.1.1 Support Vector Machine

Support Vector Machine is one of the most popular machine learning algorithms. SVM is based on the non-probabilistic binary classifier and is considered linear or non-linear, SVM is known for its effectiveness in both classification and regression problems, particularly in high-dimensional spaces.. When given a set of training inputs, each labelled with a class, The SVM builds a model around the training input and assigns new inputs to one of the classes. This makes SVM a probabilistic classifier, which is a classifier that doesn't only classify based on the most likely class, it gives a classification that is meaningful and useful on its own. Figure 2-2.

2.2.3.1.2 Unsupervised machine learning

Unsupervised learning algorithms [17] differ from the algorithms of supervised learning in that data is entered without output, and the data is not labelled and is not treated based on prediction or classification. Instead, it is treated on a clustering basis, whereby data that are similar or similar in terms of content are included in the groups. In addition, the accuracy of the output is unpredictable and often inaccurate. An example of this is the k-means algorithm. Based on what we mentioned earlier, the unsupervised learning method is where the data is not labelled, so the data cannot be divided into training and testing, we will use SSIM algorithm and supervised machine learning in our system.

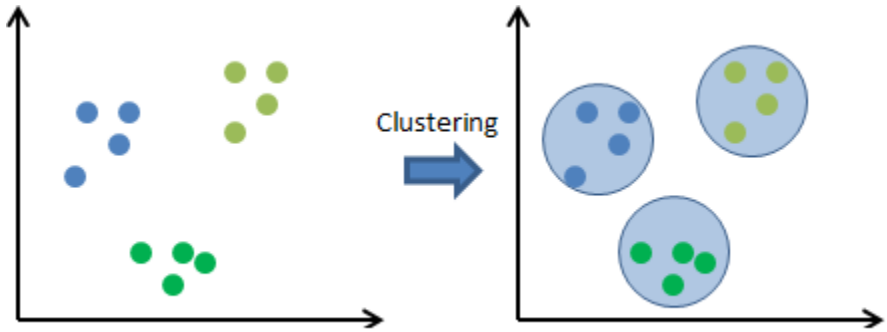


Figure 2-3 Clustering [18].

2.2.3.1.2.1 Structural Similarity Index

The Structural Similarity Index (SSIM) is a metric used to assess the quality of an image by comparing it to a reference image, which is typically of high quality. This quantitative index evaluates the similarity between two images based on three key factors: luminance, contrast, and structural information. The SSIM value is derived from the analysis of these parameters to determine the level of quality or similarity of the compared image to the reference [19]. To determine the difference between the two face sides we will use it in our system.

2.2.3.2 Deep Learning

Deep Deep learning is a subset of machine learning that involves neural networks with three or more layers. These neural networks attempt to simulate the behaviour of the human brain to "learn" from large amounts of data. The "deep" in deep learning comes from the multiple layers through which the data is transformed. It's a key technology behind artificial intelligence (AI) and applications like image and speech recognition, natural language processing, and many other areas [20].

We use deep learning in our project because the iterative nature of deep learning allows for continuous improvement of the system's performance over time. As more data is collected and new insights are gained, the models can be updated and refined, enhancing the accuracy and effectiveness of the facial and eye addiction recognition

system. So Deep learning will play a crucial role in developing and implementing the facial and eye addiction recognition system.

2.2.3.2.1 Convolutional Neural Network

The Convolutional Neural Network (CNN) is one of the most widely used deep neural networks. CNN comprises three main types of neural layers, namely, (1) convolutional layers, (2) pooling layers, and (3) fully connected layers. Each type of layer plays a different role. CNNs have been extremely successful in computer vision, especially with image data [21].

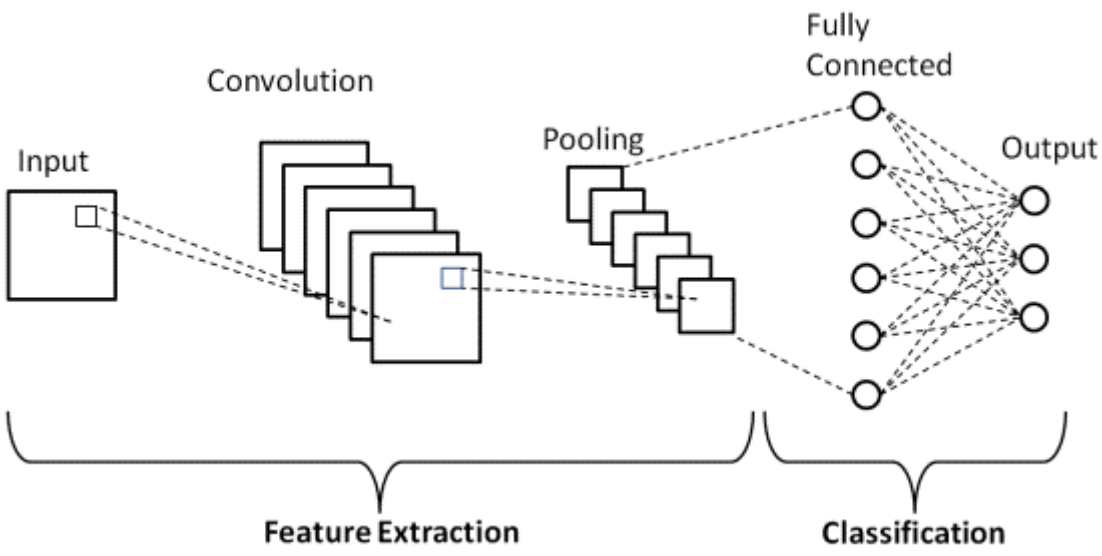


Figure 2-4 Convolutional Neural Network Layers *[22]*

2.3 Related Work

2.3.1 Review of Relevant Work

The author [23] Presented in the paper to generate a facial drug abuse image using a genetics algorithm (3DG-GA), a method for generating synthetic faces exhibiting drug abuse

traits. The study focuses on the Illicit Drug Addicts database (IDAD) containing images of individuals addicted to methamphetamine to create a realistic synthetic database while avoiding legal or ethical restrictions. The result of the study is the generation of a synthetic database of highly realistic and anonymous faces exhibiting drug abuse traits. The scientific community can use the synthetic database.

In [11] the authors produced research paper focuses on the development of a framework for the detection of illicit drug abusers through the analysis of facial images. The paper discusses the growing number of illicit drug abusers globally and the physiological side effects of illicit drug abuse, which include noticeable changes in the face. It emphasizes the significance of detecting illicit drug abusers for job or army recruitment applications, such as background verification. The proposed framework combines a Deep Convolutional Neural Network (DCNN) with a Support Vector Machine (SVM) classifier to identify illicit drug abusers' faces. A dataset of 221 illicit drug abusers' facial images, along with 221 non-abusers images, is created for experimentation. The study reports that the proposed model achieved an accuracy of 98.5% in predicting whether a person in an image is an illicit drug abuser. This paper is about using deep learning to detect if a person is an illicit drug abuser or not by building a model that can identify them accurately just by using their facial photos. Two datasets are used in this paper. One is for the Illicit Drug Abused Face dataset containing 221 images. The other is for Random facial image non-abusers from the (LFW, n.d.) database. The authors compare accuracies obtained from the various hand-crafted descriptors used with various classifiers, and the proposed method has been done. Then, the authors perform the following experiments: used basic hand-crafted representations such as Histogram of Oriented Gradients (HOG), Gastrointestinal stromal tumors (GIST), local Binary Pattern (LBP), Binarized Statistical Image Feature (BISF), and Bag of Word (BOWs) to get the best possible facial features and the classifiers RF, SVM and KNN are trained with all the representations, respectively. Proposes a deep learning framework using two CNNs - one to extract facial features and other salient features. These are concatenated and fed to an SVM classifier. Fine-tune the CNNs on the datasets using techniques like dropout and momentum to learn better representations for classification.

Technique	Accuracy(%)
HOG+SVM	89.57
GIST+SVM	94.78
BSIF+SVM	72.17
LBP+SVM	91.30
BagOfWords+SVM	87.96
HOG+Random Forests	91.30
GIST+Random Forests	93.91
BSIF+Random Forests	89.57
LBP+Random Forests	93.91
BagOfWords+Random Forests	92.59
HOG+KNN	84.35
GIST+KNN	94.78
BSIF+KNN	74.78
LBP+KNN	93.91
BagOfWords+KNN	83.33
Proposed Approach	**98.5**

Figure 2-5 The Technique Result in [11]

In [24] The researcher proposed the impact of illicit drug abuse, specifically methamphetamine .on face symmetry using state-of-the-art tools for local and global 2D face symmetry assessment, It compares the asymmetry of faces across temporal variations in meth addicts to that of ordinary individuals ageing, using two specifically collected databases. The methodology employs credible tools for local and global 2D methods, including bilateral-based metrics, the Area Mismatch metric, and the Structural Similarity Index (SSIM) for textural facial asymmetry assessment. finding a significant increase in facial asymmetry for meth addicts compared to ordinary individuals, three to five times more. the result for ordinary people the symmetry changes from 7% to 8% in the geometrical asymmetry index, and in the textural asymmetry index from 26% to 29%. The result for illicit drug users in geometric asymmetry index from 8% to 13% and in Textural asymmetry index from 28% to 39%.

This study [25] proposes a novel method to detect PDUD using facial images. Various experiments are designed to obtain the convolutional neural network (CNN) model by transfer learning based on a large-scale dataset (9870 images from PDUD and 19,567

images from GP (the general population)). Our results show that the model achieved 84.68%, 87.93%, and 83.01% in accuracy, sensitivity, and specificity in the dataset, respectively. in different facial areas. Compared with GP, the facial features of PDUD were concentrated in the areas of nose (42.98%), left cheek (44.91%), and right cheek (44.85%), which also reveals the potential relationship between mechanisms of drug action and changes in facial. tissues. This is the first study to apply the CNN model to screen PDUD in clinical practice and is also the first attempt to quantitatively analyze the facial features of PDUD. This model could be quickly integrated into the existing clinical workflow and medical care to provide capabilities.

In [26] the paper proposed approach for drug abuse detection using facial biometrics , using facial biometrics to detect prolonged drug abuse, focusing on the signs and symptoms of methamphetamine and heroin usage.by proposes a two-stage approach integrating machine learning and probabilistic reasoning to identify "drug affect facial attributes" and analyze them using a Bayesian network for drug addiction risk assessment. The study uses pre-trained convolutional neural networks (CNNs) such as GoogleNet, ResNet50, and VGG16 for face attribute classification. The proposed approach achieved an average accuracy of 90% using the ResNet50 model. the average accuracy of 84%.

Raghavendra Ramachandra et.al [27], proposed the paper Improving the ability of the facial recognition system to deal with changes resulting from illicit drug use. Dealing with the challenges of facial recognition in cases of persistent drug abuse and the effects this can have on facial geometric and textural properties. The article presents a new system based on the joint representation of the effects of independent statistical filters computed on facial images taken before and after drug use. And they only analyzed facial features and obtained an accuracy of 52%.

In [13] authors examine the pupillary light reflex (PLR) as a diagnostic tool for eye diseases and nervous system conditions, using pupil size as a parameter for predicting a patient's physical condition. Based on a convolutional neural network (CNN) for real-time pupil size prediction in a low-cost mobile embedded system. Furthermore, it explores the relevance of pupil size in psychology, indicating that pupil dilation is associated with a pleasant mood, and thus proposes an AI-based algorithm for efficient and low-cost pupil size

measurement. It recommends a model based on a CNN with specific filter types and dilation rates, providing a mean error of 5.437% for pupil length and 10.57% for pupil area, and it is achieved high accuracy in predicting pupil size, 94.56% for pupil length and 95.23% for pupil area.

Table 2-1 Summary of the related work

Paper ID	Objective	Dataset	Algorithm	Accuracy
[16]	The study aimed to create a realistic and anonymous synthetic database of facial images showing traits of drug abuse	Initially dataset of 120 images. the 3DG-GA containing 3000 synthetic facial drug abuse images	▪ 3DG-GA ▪ DEDNet	the structure for the fake image closed to the real image
[2]	In this research paper, a framework is proposed for the detection of an illicit drug abuser's face. The objective of the framework is to identify if a person is an illicit drug abuser solely by analyzing their facial image.	dataset of 221 illicit drug abusers' facial images and 221 non-abuser's facial images from LFW database	▪ DCNN ▪ SVM	98.5% For DCNN and SVM
[17]	Eth addicts have a three to five times higher rate of facial asymmetry than the general population as they age 1. The study also concluded that there is a significant increment of facial asymmetry for meth addicts while aging.	FERET database that contain 1196 and 120 members for illicit drugs abuser	▪ Geometrical asymmetry assessment: 1. Bilateral features 2. Area mismatch ▪ Textural asymmetry assessment	Geometrical asymmetry the FERET increased by 1% and the illicit abuser drugs increased by 5% In Textural asymmetry the FERET increased by 3% and the illicit addiction increased by 11%
[18]	The paper proposes the use of facial feature detection technology utilizing convolutional neural networks (CNN) to assist in the clinical screening of individuals with drug use disorders	total of 51,677 images.	▪ VGG-19, ResNet-18, and Inception. ▪ stochastic gradient descent (SGD) and adaptive moment estimation (Adam).	VGG-19 + Adam: 51.02% - VGG-19 + SGD: 61.23% - Inception + Adam: 54.99% - Inception + SGD: 79.13% - Resnet-18 + Adam: 71.61% - Resnet-18 + SGD: 79.72%
[19]	addresses the problem of detecting the prolonged drug abuse marks using "soft" face biometrics (blisters, tissue loss, hair loss, poor skin tone) by integrates both the machine learning approach and the probabilistic reasoning	112 images. .	▪ pre-trained model ResNet50 and SVM ▪ Bayesian network ▪ CNN and SVM	▪ 80% ▪ 84% ▪ 66%
[20]	improving the ability of the facial recognition system to deal with changes resulting from illicit drug use. Dealing with the challenges of facial recognition in cases of persistent drug abuse and the effects this can have on facial geometric and textural properties.	Illicit Drug Abuse 100 images of subjects.	▪ BSIF for image processing ▪ Pro-CRC for classification	They only analyzed facial features and obtained an accuracy of 52%
[21]	The paper discusses the development of a convolutional neural network (CNN) model for predicting pupil size accurately. The proposed approach achieved high accuracy in predicting pupil size, which can have significant applications in various fields.	labeled as pupils in the wild (LPW), CASIA-Iris, and ´Swirski datasets	▪ convolutional neural network (CNN)	94.56% for pupil length and 95.23% for pupil area

2.3.2 Relationship Between the Relevant Work and Our Own Work

Table 2-2 The Relationship Between the Relevant Work and Our Own Work

Name	Classification	
	Eye	**Face**
generation of artificial facial drug abuse images using deep de-identified anonymous dataset augmentation through genetics algorithm (3dg-ga)	No	Yes
Deep Learning Framework for Detection of an Illicit Drug Abuser Using Facial Image	No	Yes
Methamphetamine drug abuse and addiction: Effects on face asymmetry.	No	Yes
A Method for Detecting and Analyzing Facial Features of People with Drug Use Disorders	No	Yes
Face Attributes and Detection of Drug Addicts	No	Yes
Collaborative Representation of Statistically Independent Filters' Response: An Application to Face Recognition under Illicit Drug Abuse Alterations	No	Yes
Pupil Size Prediction Techniques Based on Convolution Neural Network	Yes	No
Our system	Yes	Yes

2.4 Summary

In this chapter, we discussed systems like the idea of DD, with a simplified explanation of each system. We also created a comparison table of all the features in each system and mentioned the methods, Pros, Cons, and accuracy.

Chapter 3: System Analysis

3.1 Introduction

As we mentioned earlier, we are seeking to develop a system that helps detect drug addiction with the least amount of time, effort, and money just by taking a picture of the suspected person. So, in this chapter, we will collect and interpret facts, identify the problems, and decompose a system into its components to identify its objectives and ensure that all the components of the system work efficiently to accomplish their purpose.

3.2 Requirements Elicitation

The Requirement Elicitation is a fundamental stage in the system development life cycle where the focus is on understanding and defining the application domain and services the system should provide.

3.2.1 Functional Requirements

3.2.1.1 Functional requirements related to user.

The users authorized to use our system are members of the general directorate of Narcotics Control.

- FR1: The user shall be able to login.
- FR2: The user shall be able to open the camera by OpenCV.
- FR2.1: The user should be able to take pictures of eyes or face.
- FR3: The user should be able to view the result.

3.2.1.2 Functional requirements related to the system.

- FR1: The system shall be able to obtain a username and password from the user.
- FR2: The system should redirect the user from the login page to the camera page.
- FR3: The system should be able to classify the image and decide if this image belongs to the abuser or a normal person.

- FR4: The system should be able to display the result.

3.2.2 Non-Functional Requirements

- Performance: The system should be designed to process images and perform analysis within acceptable time frames
- Accuracy: The system should achieve a minimum accuracy of 80% in correctly classifying image.
- Scalability: The system should be scalable to accommodate many images captures and analyses.
- Usability: The user interface should be intuitive and user-friendly.

3.3 Requirements Specification

The software requirements specification describes the proposed systems' functional requirements. There are several methods to understand the system's actions and how they are. Communicates with its users to satisfy their needs; one of them is UML diagrams. The following section explains the system requirements using a UML use case diagram.

3.3.1 Use Case Diagram

The use case diagram is one of the most significant diagrams that helps understand the system's functionality. It describes the interaction between the user and the system from the user’s point of view. The system use case is shown in:

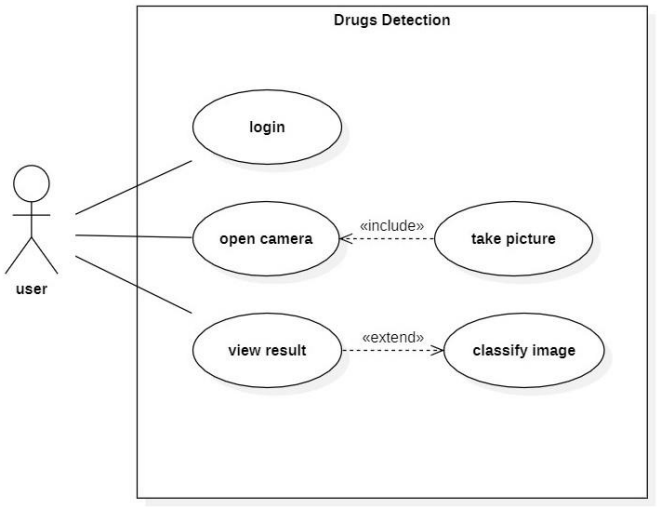


Figure 3-1 Use Case

3.3.2 Description of use case

Table 3-1 Classify Image Use Case Description

Use case name	Classify image
Participating actor	System
Description	The system shall be able to classify the captured images based on predefined categories (normal, drug abuser) related to facial or eye characteristics.
Pre – condition	The user should be able to take pictures for eyes or face.
Post condition	The system shall be able to display the results.
Flow of event	1. User will open camera. 2. User will take picture. 3. Sends the captured image to the model. 4. The model will classify image. 5. The model will display the results to the user.

Table 3-2 Take Picture Description

Use case name	Take a picture
Participating actor	User
Include use case	Analys the picture
Description	The user takes a picture for face or eye
Pre – condition	The user must log in

Post condition	system display the analysed picture
Flow of event	1. User will open the application. 2. User will login. 3. User will open camera. 4. User can take a picture.

Table 3-3 Login Use Case Description

Use case name	Log in
Participating actor	User
Include use case	Authentication
Description	The system shall provide a dedicated login page accessible to users.
Pre – condition	The user has a valid account credentials (username and password).
Post condition	The user is now logged in and has access to their account features.
Flow of event	1. User opens the application. 2. The user is presented with a welcome screen with a login option. 3. Users choose the login option. 4. Users enter their username and password in the provided fields. 5. User press login option. 6. If the credentials are valid, redirect to take picture page. 7. If the credentials are invalid, the system displays an error message indicating the invalid credentials and prompts the user to retry.

3.4 Research methodology

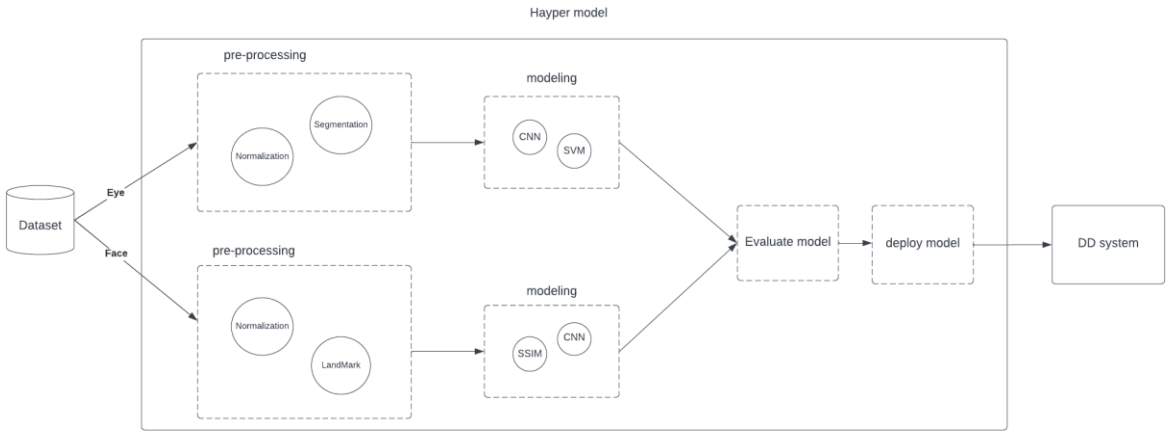


Figure 3-2 Research Methodology Chart

3.4.1 Proposed Model:

Figure 3-2 introduces an overview of the Methodology of Drug Detection (DD) Models workflow. It consists of Two Parts, Detecting by (Eyes or Face). All these parts have the same stages but different algorithms.

3.4.2 Dataset Collection:

We collect data sets containing eye and face images of normal people and people who use stimulants. We had difficulty collecting images due to patient privacy, but with intensified research and activation of the VPN, we were able to collect a good number of images.

3.4.3 Eyes:

3.4.3.1 Preprocessing:

To increase the accuracy of the eyes image we will use two phases, first we will be normalizing the image such as (rotate, translate, scale, resize, convert image to gray) to get image in right. position. Then we will use segmentation to improve the image and cropping the pupils.

3.4.3.2 Modeling:

Once the image is improved, it will be passed into the model we will try a different approach. We will try a different approach. to build the model, one of these is using CNN and SVM.

3.4.4 Face:

3.4.4.1 Preprocessing:

The face must be of the right size and aligned to provide comparable results, therefore we will. use (rotate, translate, scale, resize, convert image to gray) to have the face in the boundaries what we are looking for. Then we will use LandMark algorithm, which extracts the face in 68 points, to create detailed map of facial features.

3.4.4.2 Modeling:

After extracting the face feature, we will use different approaches such as CNN and SSIM, trained a CNN model to has rich facial information that helps to determine the drug abused, the SSIM we will be cropping the face and compere the right side with the left side it. abused, the SSIM we will be cropping the face and compare the right side with the left side it will return value between 0 and 1 which is 1 indicate perfect similarity and 0 indicates to no similarity.

3.4.5 Evaluate Model:

Once the model is trained, we will evaluate the performance by using appropriate evaluation metrics, such as (accuracy, Confusion matrix, recall, F1 score, and ROC curve).

3.4.6 Deploy Model:

We will deploy the model in a controlled environment to assess its usability, reliability, and accuracy, and make necessary adjustments or refinements to improve the system.

3.5 Developmental methodology

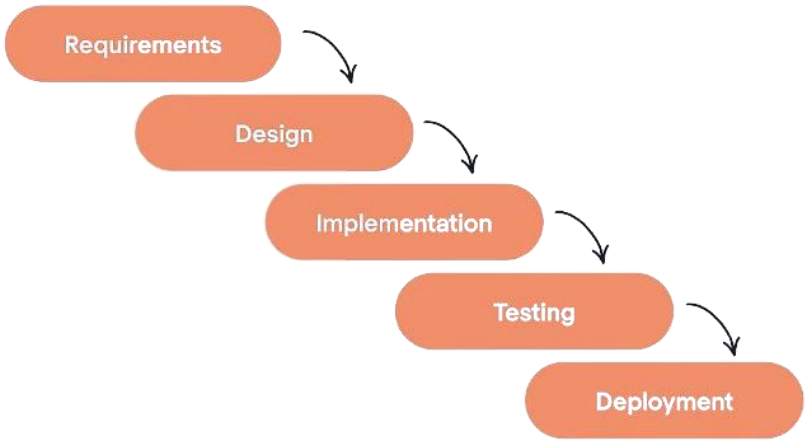


Figure 3-3 Waterfall Model [28]

In the context of DD, the phases of the waterfall model are applied as follows:

1- **Requirements**: We spent a lot of effort reading papers close to our idea and gathering the databases we needed to make the requirements a solid foundation for the later stages of our project.
2- **Design**: During this phase, we defined the base components and architecture of the system by using multiple diagrams such as use case and class diagrams. Also, we used Figma prototyping tools to design the UI of our application.
3- **Implementation**: During this phase, we use the inputs of the previous phase to develop the system and its functions as modules, which will be tested and integrated in the next phase. We expect to use Flutter and Python to implement the system.
4- **testing**: During this phase, all modules developed in the implementation phase will be tested to ensure that there are no errors. After testing each module, we will integrate it into the system and make sure it works as a complete system.
5- **Deployment**: After completing testing of the entire system, the application will be deployed in the client's environment.

3.6 Summery

In this chapter, we presented the requirements of the system. Also, a use case diagram that explains the interaction of the system very precisely is introduced, followed by descriptive tables that show the process of each use case.

In the end, we describe the research methodology and development methodology of the system, which explains how we collect information, how we applied for this work, and what constraints we encountered.

Chapter 4: System Design

4.1 Introduction

System design is a crucial phase in the software development life cycle. Therefore, we will ensure our Drug Detection (DD) system encompasses the creation of an architecture that fulfils the specified requirements and ensures the efficient and effective functioning of the proposed system.

4.2 Architectural Design

Architecture design provides an abstraction to manage system complexity and establish communication and coordination mechanisms among components.

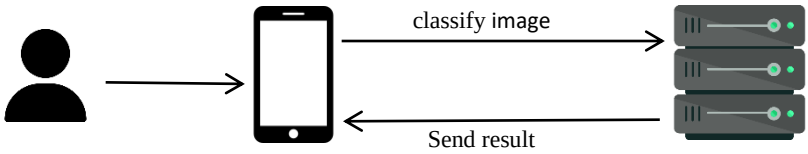


Figure 4-1 Architectural Design

In figure 4-1 we choose client to server approach it is partitions tasks between the providers of a service, called **servers,** and service requesters, called **clients** [29]**.**

In our system, the user interacts with the front end. This level contains the login screens, take pictures, and result that are given to user. Briefly, any information is given to the user or taken from the user.

After that comes the turn of the back-end, which the server takes the image and passes it to the model to classify it and sends the result to the client using the front-end module.

4.3 Object Oriented Design

Object-oriented design is concerned with defining the modules in the system and how they interact with each other. In this section, we model our system by flowchart and using UML language sequence diagram, and class diagram, and we also show the user interface design of the system.

4.3.1 Structural Static Models

The class diagram explains the structures of our system by determining attributes, operations, and relationships between objects. Shows the class diagram of the system. Flowchart diagrams are used to model the flow of events and the different states the system goes through.

- Flowchart :

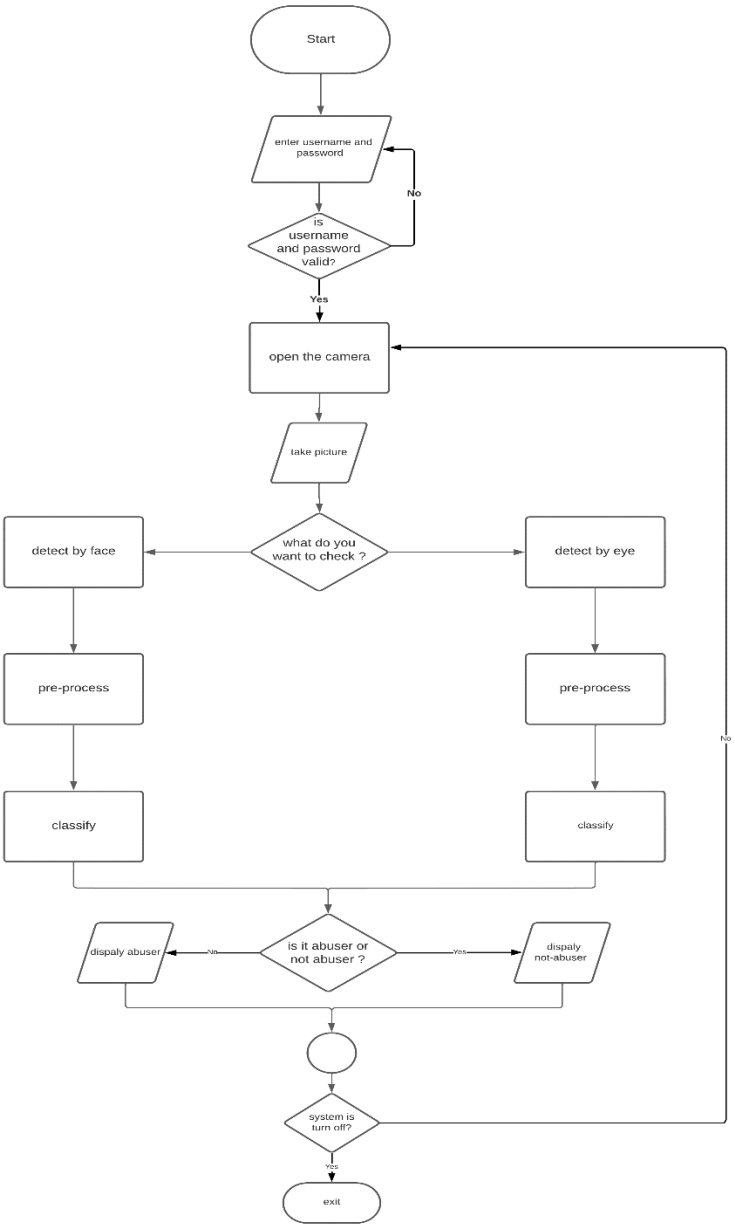


Figure 4-2 Flowchart

- Class diagram :

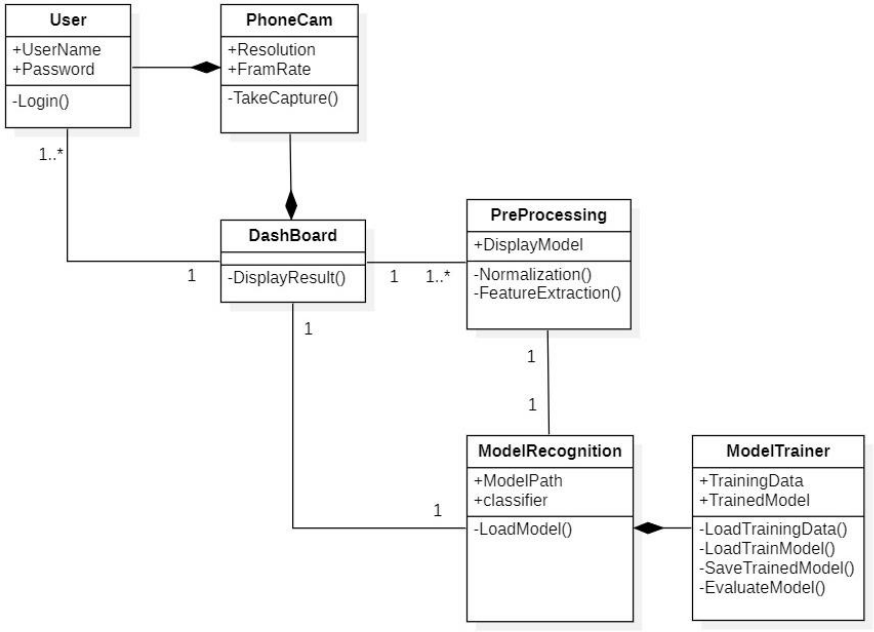


Figure 4-3 Class diagram

4.3.2 Dynamic Models

This section describes how the system's actors and objects interact with each other over a period and describes the basic dynamic sequence diagrams for the system. sequence diagram:

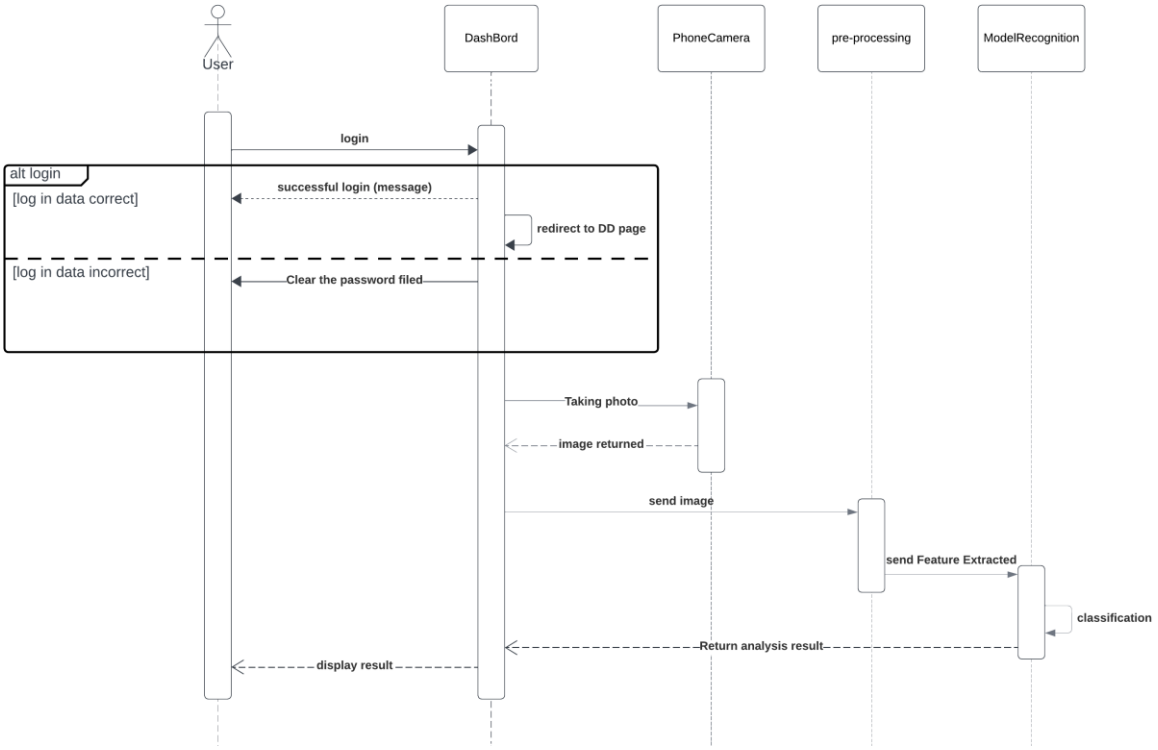


Figure 4-4 Sequence Diagram.

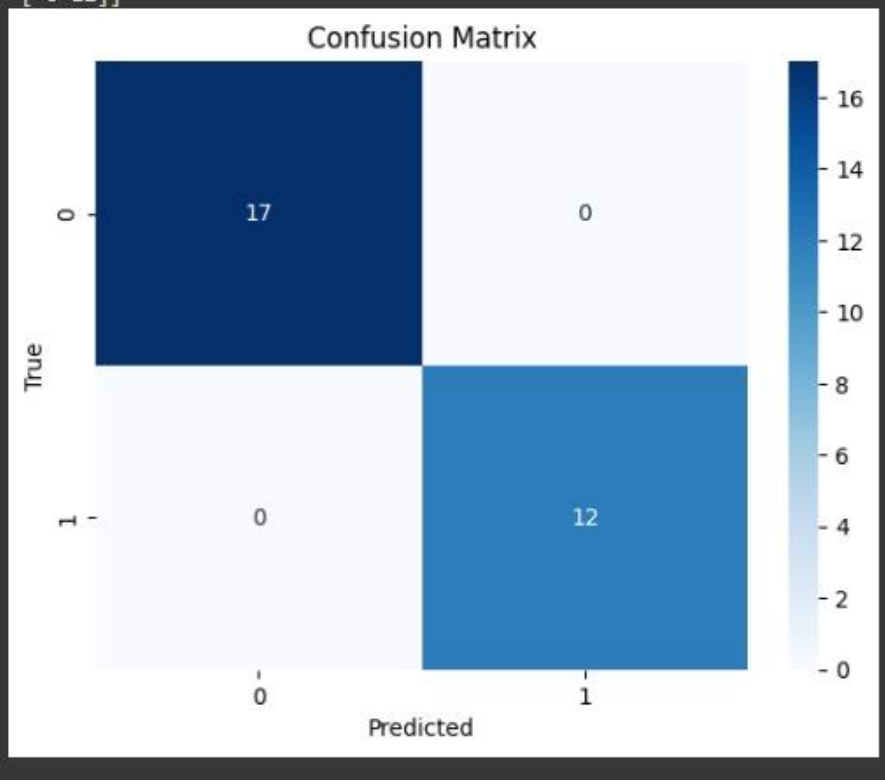
4.3.3 Model

```
[12] import cv2
     import numpy as np
     import os
     from sklearn.model_selection import train_test_split
     # Paths data folders
     dataset_path = '/content/drive/MyDrive/databases'

     data_meth_path = os.path.join(dataset_path, '/content/drive/MyDrive/databaes/meth')
     data_normal_path = os.path.join(dataset_path, '/content/drive/MyDrive/databaes/normal')

     # Function to load and preprocess data
     def load_data(data_path, label):
         image_files = [f for f in os.listdir(data_path) if os.path.isfile(os.path.join(data_path, f))]
         images = []
         labels = []
         for image_file in image_files:
             img = cv2.imread(os.path.join(data_path, image_file), cv2.IMREAD_GRAYSCALE)
             img = cv2.resize(img, (200, 200))
             images.append(img)
             labels.append(label)
         return np.array(images), np.array(labels)

     # Loading data
     X_meth, y_meth = load_data(data_meth_path, 0)  # Label 'meth' as 0
     X_normal, y_normal = load_data(data_normal_path, 1)  # Label 'normal' as 1

     # Combining and splitting data
     X = np.concatenate((X_meth, X_normal), axis=0)
     y = np.concatenate((y_meth, y_normal), axis=0)
     X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
from sklearn import svm
from sklearn.metrics import accuracy_score
import joblib


# Reshaping data for SVM
X_train = X_train.reshape(X_train.shape[0], -1)
X_test = X_test.reshape(X_test.shape[0], -1)

# Training SVM classifier
classifier = svm.SVC(kernel='linear')
classifier.fit(X_train, y_train)

# Evaluating the classifier
y_pred = classifier.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy}')

# Saving the model
joblib.dump(classifier, os.path.join('/content/drive/MyDrive/databaes/theclass', 'classifier.pkl'))
```

```python
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

# Visualize the confusion matrix
sns.heatmap(cm, annot=True, fmt="d", cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.title('Confusion Matrix')
plt.show()
```

```
Confusion Matrix:
[[17  0]
 [ 0 12]]
```



Demo.mp4

4.4 User Interface Design

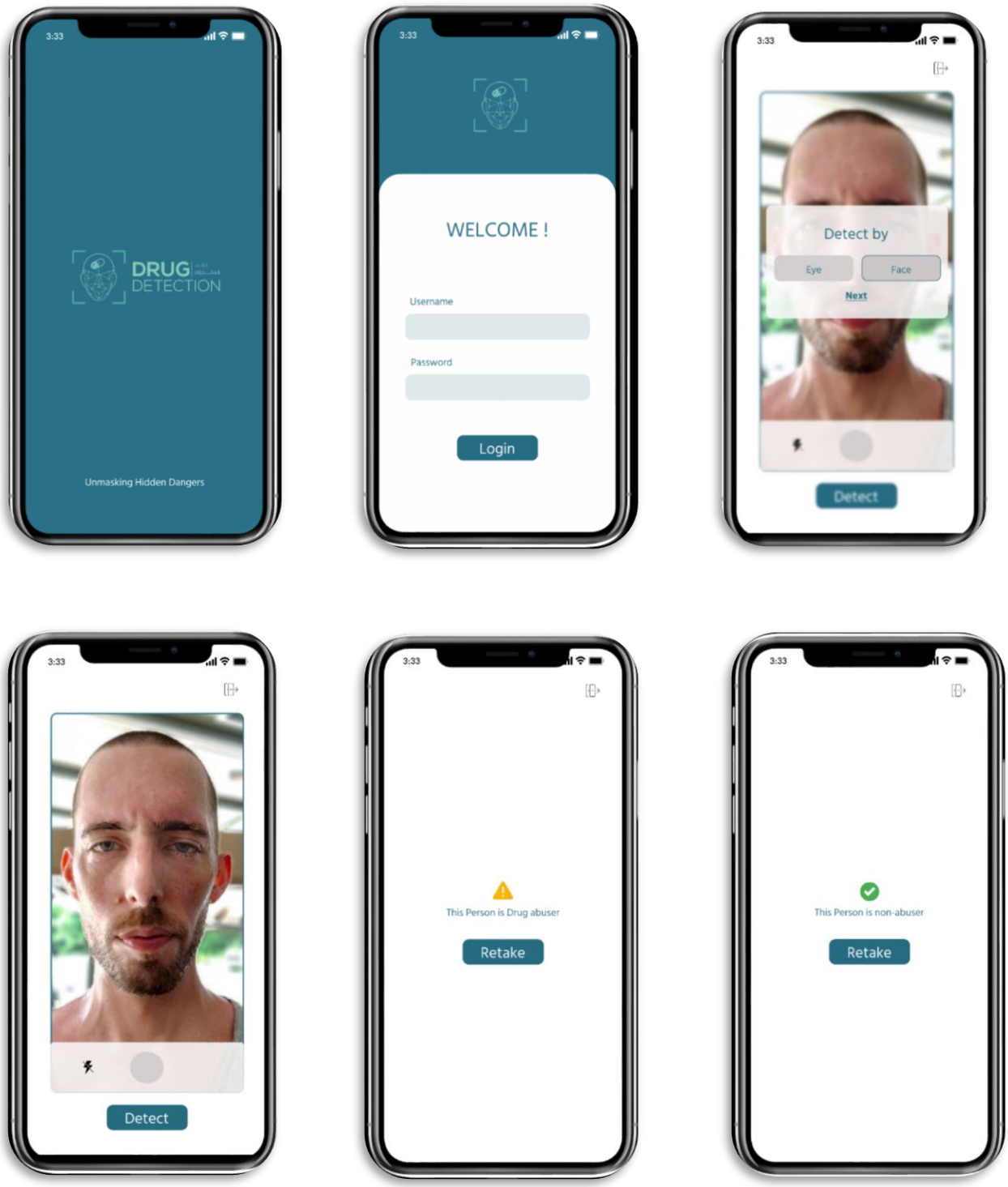


Figure 4-5 User Interface

4.5 Summary

In this chapter, we described the overall organization of the system from different aspects. We discussed the architectural design, user interface, and system diagrams, such as Flowchart and sequence diagrams.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

In conclusion, the issue of drug addiction presents a significant health and social challenge in contemporary society. This addiction not only induces detrimental physiological alterations in the human body but also pupils in the eyes and facial features. Critical indicators of drug use include changes in pupil size and facial appearance. in regular pupil sizes in healthy adults vary between 2 to 4 mm in bright light and 4 to 8 mm in darkness. In contrast, drug users often show dilated pupils that do not constrict appropriately in response to light variations. Furthermore, the addiction to illegal stimulants like methamphetamine and cocaine leads to clear pupil dilation and can accelerate facial ageing, causing noticeable deformities and asymmetries.

To combat this pervasive issue, the implementation of the drug detection (DD) system is imperative. This system leverages advanced deep learning and machine learning methodologies, including convolutional neural networks (CNNs), support vector machines (SVMs), and structural similarity index measurement (SSIM) algorithms. By analyzing images of an individual's eyes and face, the system can identify signs of drug use. This creative approach not only aids in the timely identification of drug users but also plays a crucial role in preventing dangerous behaviors in individuals with impaired awareness, thereby facilitating early intervention and offering essential support. Such technological advancements are pivotal in addressing the multifaceted challenges posed by drug addiction in modern society.

5.2 Goals Achieved

We have studied the work properly during this semester. Thus, we achieved the following outcomes:

- Document the requirements that are required for the work after gathering and analyzing the needs of the project.
- Completing the design of the system components and structure.

5.3 Limitations and Future Work

Certain limitations should be considered, such as the system's performance being heavily reliant on the quality and diversity of the training dataset. Insufficient or biased data may result in reduced accuracy of the system's predictions. Therefore, it was not easy to focus on expanding and diversifying the dataset to improve the system's robustness.

In future work, we will start to implement, evaluate, and lastly deploy our system to contribute to controlling drug addiction.

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